# CMPSC 465, Fall 2025
# Homework 1

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## 1. Getting Started

I understand the course policies.

## 2. Comparing Growth Rates

*For each pair, we determine the asymptotic relationship by analyzing the dominant terms, often using the limit test: Let* $L = \lim_{n\to\infty} \frac{f(n)}{g(n)}$. *If* $L = 0$, $f = O(g)$. *If* $L = \infty$, $f = \Omega(g)$. *If* $L$ *is a positive constant,* $f = \Theta(g)$.

**(a)** $f(n) = 4n \cdot 3^n + n^{50}, \quad g(n) = 4^n$

The dominant term in $f(n)$ is $4n \cdot 3^n$. We compare it to $g(n)$.

$$\lim_{n\to\infty} \frac{4n \cdot 3^n}{4^n} = \lim_{n\to\infty} 4n \left(\frac{3}{4}\right)^n = \lim_{n\to\infty} \frac{4n}{(4/3)^n}$$

This is an indeterminate form $\frac{\infty}{\infty}$, so we apply L'Hôpital's Rule:

$$\lim_{n\to\infty} \frac{4}{(4/3)^n \ln(4/3)} = 0$$

Since the limit is 0, $f(n)$ grows strictly slower than $g(n)$. **Answer:** $f = O(g)$

**(b)** $f(n) = \log(5n), \quad g(n) = \log(4n)$

Using logarithm properties, $f(n) = \log 5 + \log n$ and $g(n) = \log 4 + \log n$.

$$\lim_{n\to\infty} \frac{\log 5 + \log n}{\log 4 + \log n} = \lim_{n\to\infty} \frac{(\log 5/\log n) + 1}{(\log 4/\log n) + 1} = \frac{0+1}{0+1} = 1$$

Since the limit is a positive constant, the functions grow at the same rate. **Answer:** $f = \Theta(g)$

**(c)** $f(n) = e^n + n^{10}, \quad g(n) = 2^n$

The dominant term in $f(n)$ is $e^n$. We compare it to $g(n)$. Since $e \approx 2.718 > 2$, we expect $f(n)$ to grow faster.

$$\lim_{n\to\infty} \frac{e^n}{2^n} = \lim_{n\to\infty} \left(\frac{e}{2}\right)^n = \infty$$

Since the limit is $\infty$, $f(n)$ grows strictly faster than $g(n)$. **Answer:** $f = \Omega(g)$

**(d)** $f(n) = 7^n, \quad g(n) = 8^n$

We compare the two exponential functions.

$$\lim_{n\to\infty} \frac{7^n}{8^n} = \lim_{n\to\infty} \left(\frac{7}{8}\right)^n = 0$$

Since the limit is 0, $f(n)$ grows strictly slower than $g(n)$. **Answer:** $f = O(g)$

**(e)** $f(n) = \log(n^8 + n^2), \quad g(n) = \log(6n)$

For large $n$, $f(n) = \log(n^8(1 + n^{-6})) = 8\log n + \log(1 + n^{-6}) \approx 8\log n$.

$$\lim_{n\to\infty} \frac{8\log n + \log(1+n^{-6})}{\log 6 + \log n} = \lim_{n\to\infty} \frac{8 + \frac{\log(1+n^{-6})}{\log n}}{\frac{\log 6}{\log n} + 1} = \frac{8+0}{0+1} = 8$$

Since the limit is a positive constant, the functions grow at the same rate. **Answer:** $f = \Theta(g)$

**(f)** $f(n) = 2^n + n^{100}$, $\quad g(n) = 3 \cdot 2^n + \log n$

The dominant terms are $2^n$ in $f(n)$ and $3 \cdot 2^n$ in $g(n)$.

$$\lim_{n\to\infty} \frac{2^n + n^{100}}{3 \cdot 2^n + \log n} = \lim_{n\to\infty} \frac{1 + n^{100}/2^n}{3 + (\log n)/2^n} = \frac{1+0}{3+0} = \frac{1}{3}$$

Since the limit is a positive constant, the functions grow at the same rate. **Answer:** $f = \Theta(g)$

**(g)** $f(n) = \sqrt{n} + \log n$, $\quad g(n) = 4\sqrt{n} + n^{1/3}$

The dominant terms are $\sqrt{n} = n^{1/2}$ in $f(n)$ and $4\sqrt{n}$ in $g(n)$.

$$\lim_{n\to\infty} \frac{\sqrt{n} + \log n}{4\sqrt{n} + n^{1/3}} = \lim_{n\to\infty} \frac{1 + (\log n)/\sqrt{n}}{4 + n^{1/3}/\sqrt{n}} = \frac{1+0}{4+0} = \frac{1}{4}$$

Since the limit is a positive constant, the functions grow at the same rate. **Answer:** $f = \Theta(g)$

**(h)** $f(n) = (\log_3 n)^{\log_3 n}$, $\quad g(n) = 3^{(\log_3 n)^2}$

We rewrite both functions as powers of 3. Let $x = \log_3 n$. $g(n) = 3^{x^2} = 3^{x \cdot x} = 3^{(\log_3 n)(\log_3 n)}$. $f(n) = x^x = 3^{\log_3(x^x)} = 3^{x \log_3 x} = 3^{(\log_3 n)(\log_3(\log_3 n))}$. We compare the exponents. For large $n$, $\log_3(\log_3 n) < \log_3 n$. Therefore, the exponent of $f(n)$ is strictly smaller than the exponent of $g(n)$. **Answer:** $f = O(g)$

**(i)** $f(n) = \frac{\log n}{n}$, $\quad g(n) = n^{1/n}$

First, analyze $g(n)$. Let $y = n^{1/n}$. Then $\ln y = \frac{\ln n}{n}$. As $n \to \infty$, $\lim \frac{\ln n}{n} = 0$ by L'Hôpital's Rule. Thus, $\lim y = e^0 = 1$. So $g(n)$ approaches 1. For $f(n)$, $\lim_{n\to\infty} \frac{\log n}{n} = 0$. Since $f(n) \to 0$ and $g(n) \to 1$, $f(n)$ grows strictly slower than $g(n)$. **Answer:** $f = O(g)$

## 3. Proofs

### (a) Prove that $3n^2 + 4n + 5 = \Theta(n^2)$

To prove $f(n) = 3n^2 + 4n + 5 = \Theta(n^2)$, we must find positive constants $c_1, c_2, n_0$ such that $c_1 n^2 \leq 3n^2 + 4n + 5 \leq c_2 n^2$ for all $n \geq n_0$.

- **Upper Bound ($O$):** We want to show $3n^2 + 4n + 5 \leq c_2 n^2$. For $n \geq 1$, we know $4n \leq 4n^2$ and $5 \leq 5n^2$. Thus,
$$3n^2 + 4n + 5 \leq 3n^2 + 4n^2 + 5n^2 = 12n^2$$
This holds for $c_2 = 12$ and $n_0 = 1$.
- **Lower Bound ($\Omega$):** We want to show $3n^2 + 4n + 5 \geq c_1 n^2$. Since $4n > 0$ and $5 > 0$ for $n \geq 1$,
$$3n^2 + 4n + 5 \geq 3n^2$$
This holds for $c_1 = 3$ and $n_0 = 1$.

Since we found such constants, $3n^2 + 4n + 5 = \Theta(n^2)$.

### (b) Prove that if $f_1(n) = O(g_1(n))$ and $f_2(n) = O(g_2(n))$, then $f_1(n) + f_2(n) = O(\max(g_1(n), g_2(n)))$

- By definition of Big-O, there exist positive constants $c_1, n_1, c_2, n_2$ such that for all $n \geq n_1$, $|f_1(n)| \leq c_1|g_1(n)|$, and for all $n \geq n_2$, $|f_2(n)| \leq c_2|g_2(n)|$.
- Let $n_0 = \max(n_1, n_2)$ and let $g_{\max}(n) = \max(|g_1(n)|, |g_2(n)|)$. By definition of max, $|g_1(n)| \leq g_{\max}(n)$ and $|g_2(n)| \leq g_{\max}(n)$.
- For all $n \geq n_0$, we use the triangle inequality and then substitute:

  $|f_1(n)+f_2(n)| \leq |f_1(n)|+|f_2(n)| \leq c_1|g_1(n)|+c_2|g_2(n)| \leq c_1 g_{\max}(n)+c_2 g_{\max}(n) = (c_1+c_2)g_{\max}(n)$
- Let $c = c_1 + c_2$. We have found a constant $c > 0$ such that for all $n \geq n_0$, $|f_1(n) + f_2(n)| \leq c \cdot \max(|g_1(n)|, |g_2(n)|)$. This is the definition of $O(\max(g_1(n), g_2(n)))$.

### (c) Prove transitivity of Theta notation

- Given $f(n) = \Theta(g(n))$ and $g(n) = \Theta(h(n))$. By definition, there are positive constants $c_1, c_2, n_1$ such that $c_1|g(n)| \leq |f(n)| \leq c_2|g(n)|$ for $n \geq n_1$, and positive constants $c_3, c_4, n_2$ such that $c_3|h(n)| \leq |g(n)| \leq c_4|h(n)|$ for $n \geq n_2$.
- Let $n_0 = \max(n_1, n_2)$. For all $n \geq n_0$, all inequalities hold.
- **Upper Bound ($O$):** We substitute the upper bound for $g(n)$ into the upper bound for $f(n)$:
$$|f(n)| \leq c_2|g(n)| \leq c_2(c_4|h(n)|) = (c_2c_4)|h(n)|$$
This shows $f(n) = O(h(n))$ with constant $c_2c_4$.
- **Lower Bound ($\Omega$):** We substitute the lower bound for $g(n)$ into the lower bound for $f(n)$:
$$|f(n)| \geq c_1|g(n)| \geq c_1(c_3|h(n)|) = (c_1c_3)|h(n)|$$
This shows $f(n) = \Omega(h(n))$ with constant $c_1c_3$.
- Since $f(n)$ is both $O(h(n))$ and $\Omega(h(n))$, we conclude $f(n) = \Theta(h(n))$.

## 4. Useful Identities

### (a) Show that $\log(n!) = \Theta(n \log n)$

- We start with the property $\log(n!) = \sum_{i=1}^{n} \log(i)$.
- **Upper Bound ($O$):** Each term in the sum is at most $\log(n)$.

$$\sum_{i=1}^{n} \log(i) \leq \sum_{i=1}^{n} \log(n) = n \log n$$

  Thus, $\log(n!) = O(n \log n)$.

- **Lower Bound ($\Omega$):** We can lower-bound the sum by considering only the last half of the terms. The number of terms from $i = \lceil n/2 \rceil$ to $n$ is at least $n/2$. The smallest of these terms is $\log(\lceil n/2 \rceil) \geq \log(n/2)$.

$$\sum_{i=1}^{n} \log(i) \geq \sum_{i=\lceil n/2 \rceil}^{n} \log(i) \geq \sum_{i=\lceil n/2 \rceil}^{n} \log(n/2) \geq \frac{n}{2} \log\left(\frac{n}{2}\right)$$

$$\frac{n}{2}(\log n - \log 2) = \frac{1}{2} n \log n - \frac{\log 2}{2} n$$

  For $n \geq 4$, $\frac{1}{2} n \log n \geq n \log 2$, so $\frac{1}{2} n \log n - \frac{\log 2}{2} n \geq \frac{1}{4} n \log n$. This shows $\log(n!) = \Omega(n \log n)$.

### (b) Show that $\sum_{i=1}^{n} \frac{1}{i} = \Theta(\log n)$

- **Upper Bound ($O$):** We group terms by powers of two. The sum of terms in the group from $i = 2^j$ to $2^{j+1} - 1$ is bounded by the number of terms ($2^j$) times the largest term value ($1/2^j$).

$$\sum_{i=2^j}^{2^{j+1}-1} \frac{1}{i} \leq \sum_{i=2^j}^{2^{j+1}-1} \frac{1}{2^j} = 2^j \cdot \frac{1}{2^j} = 1$$

  There are approximately $\log_2 n$ such groups up to $n$. Thus, $\sum_{i=1}^{n} \frac{1}{i} \leq \sum_{j=0}^{\lfloor \log_2 n \rfloor} 1 = \lfloor \log_2 n \rfloor + 1 = O(\log n)$.

- **Lower Bound ($\Omega$):** Using the same groups, we bound the sum by the number of terms ($2^j$) times the smallest term value ($1/2^{j+1}$).

$$\sum_{i=2^j}^{2^{j+1}-1} \frac{1}{i} \geq \sum_{i=2^j}^{2^{j+1}-1} \frac{1}{2^{j+1}} = 2^j \cdot \frac{1}{2^{j+1}} = \frac{1}{2}$$

  Summing over the $\approx \log_2 n$ full groups up to $n$ gives a total sum of at least $\frac{1}{2} \log_2 n = \Omega(\log n)$.

## 5. Recurrence Relations

**(a)** $T(n) = 4T(n/2) + 5n^3$

Using the Master Theorem, we have $a = 4, b = 2, f(n) = 5n^3$. We compare $f(n)$ with $n^{\log_b a} = n^{\log_2 4} = n^2$. We see that $f(n) = 5n^3$ is polynomially larger than $n^2$. Specifically, $f(n) = \Omega(n^{2+\epsilon})$ for $\epsilon = 1$. This fits Case 3. We must check the regularity condition: $af(n/b) \leq cf(n)$ for some constant $c < 1$.

$$4 \cdot 5(n/2)^3 = 4 \cdot 5 \cdot \frac{n^3}{8} = \frac{5}{2}n^3$$

We require $\frac{5}{2}n^3 \leq c \cdot 5n^3$, which means $1/2 \leq c$. We can pick $c = 1/2$, which is less than 1. The condition holds. By Case 3, $T(n) = \Theta(f(n))$. **Answer:** $O(n^3)$

**(b)** $T(n) = 3T(n/4) + n^{0.6}$

Here, $a = 3, b = 4, f(n) = n^{0.6}$. We compare $f(n)$ with $n^{\log_b a} = n^{\log_4 3} \approx n^{0.792}$. Since $f(n) = n^{0.6}$ is polynomially smaller than $n^{0.792}$, specifically $f(n) = O(n^{\log_4 3 - \epsilon})$ for $\epsilon \approx 0.192$, this fits Case 1 of the Master Theorem. Thus, $T(n) = \Theta(n^{\log_4 3})$. **Answer:** $O(n^{\log_4 3})$

**(c)** $T(n) = 7T(n/3) + n^2$

Here, $a = 7, b = 3, f(n) = n^2$. We compare $f(n)$ with $n^{\log_b a} = n^{\log_3 7} \approx n^{1.771}$. $f(n) = n^2$ is polynomially larger than $n^{1.771}$, so we check Case 3. The regularity condition is $7 \cdot (n/3)^2 = \frac{7}{9}n^2 \leq c \cdot n^2$. This holds for $c = 7/9 < 1$. By Case 3, $T(n) = \Theta(f(n))$. **Answer:** $O(n^2)$

**(d)** $T(n) = 8T(n/2) + 3n^3$

Here, $a = 8, b = 2, f(n) = 3n^3$. We compare $f(n)$ with $n^{\log_b a} = n^{\log_2 8} = n^3$. Since $f(n) = \Theta(n^3)$, this fits Case 2 of the Master Theorem. Thus, $T(n) = \Theta(n^{\log_b a} \log n)$. **Answer:** $O(n^3 \log n)$

**(e)** $T(n) = T(n-1) + d^n$**, where** $d \geq 1$

This is solved by unrolling: $T(n) = T(n-1) + d^n = (T(n-2) + d^{n-1}) + d^n = \cdots = T(0) + \sum_{i=1}^{n} d^i$. This is a geometric series. If $d = 1$, the sum is $\sum_{i=1}^{n} 1 = n$, so $T(n) = \Theta(n)$. If $d > 1$, the sum is dominated by the largest term, $\frac{d(d^n-1)}{d-1} = \Theta(d^n)$. So $T(n) = \Theta(d^n)$. **Answer:** $O(d^n)$ **for** $d > 1$**, and** $O(n)$ **for** $d = 1$**.**

**(f)** $T(n) = T(n/3) + 0.75^n$

Here, $a = 1, b = 3, f(n) = 0.75^n$. We compare $f(n)$ with $n^{\log_b a} = n^{\log_3 1} = n^0 = 1$. The function $f(n) = (3/4)^n$ shrinks to zero as $n \to \infty$. It is bounded above by a constant, so $f(n) = O(1) = O(n^0)$. This fits Case 1. The total work is dominated by the root and base cases. **Answer:** $O(1)$

**(g)** $T(n) = T(4n/7) + T(3n/7) + \Theta(n)$

We use the substitution method as hinted. We want to prove $T(n) = O(n \log n)$. **Inductive Hypothesis:** Assume $T(k) \leq ck \log k$ for all $k < n$, for some constant $c$. Let the non-recursive work be $dn$ for some constant $d$.

$$T(n) = T(4n/7) + T(3n/7) + dn$$

Substitute the IH:

$$\begin{aligned}
T(n) &\le c\left(\frac{4n}{7}\right)\log\left(\frac{4n}{7}\right) + c\left(\frac{3n}{7}\right)\log\left(\frac{3n}{7}\right) + dn \\
&= \frac{4cn}{7}(\log n - \log(7/4)) + \frac{3cn}{7}(\log n - \log(7/3)) + dn \\
&= \frac{4cn}{7}\log n - \frac{4cn}{7}\log(7/4) + \frac{3cn}{7}\log n - \frac{3cn}{7}\log(7/3) + dn \\
&= cn\log n - \left(\frac{c}{7}(4\log(7/4) + 3\log(7/3)) - d\right)n
\end{aligned}$$

We need the entire expression to be $\le cn\log n$. This is true if the residual term is non-positive:

$$-\left(\frac{c}{7}(4\log(7/4) + 3\log(7/3)) - d\right)n \le 0$$

$$d \le \frac{c}{7}(4\log_2(7/4) + 3\log_2(7/3))$$

The term $4\log_2(7/4) + 3\log_2(7/3) \approx 4(0.807) + 3(1.222) \approx 3.228 + 3.666 = 6.894$ is a positive constant. Let's call it $K$. We need $d \le cK/7$, or $c \ge 7d/K$. Since $d$ and $K$ are positive constants, we can always choose a large enough constant $c$ to satisfy this inequality. Thus, the induction holds. **Answer:** $O(n\log n)$